Identity 3.11: $\cos 2A = \dfrac{1-\tan^2 A}{1+\tan^2 A}$

Proof.

$$\begin{aligned}\text{We know that } \cos 2A &= \cos^2 A - \sin^2 A = \frac{\cos^2 A - \sin^2 A}{\cos^2 A + \sin^2 A}\\ &= \frac{\dfrac{\cos^2 A - \sin^2 A}{\cos^2 A}}{\dfrac{\cos^2 A + \sin^2 A}{\cos^2 A}}\\ \text{Thus, } \cos 2A &= \frac{1-\tan^2 A}{1+\tan^2 A}.\end{aligned}$$

Power reducing identities or Reduction identities

Power reducing identities for sine, cosine and tangent can be derived using the double-angle identities. For example,

$$\cos 2A = 2\cos^2 A - 1 \Rightarrow \cos^2 A = \frac{1+\cos 2A}{2}$$

The following table is the list of power reducing identities.

Power Reducing Identities
$\sin^2 A = \dfrac{1-\cos 2A}{2}, \quad \cos^2 A = \dfrac{1+\cos 2A}{2}, \quad \tan^2 A = \dfrac{1-\cos 2A}{1+\cos 2A}$

(i) In the power reducing identities, we have reduced the square power on one side to power 1 on the other side.

(ii) Power reducing identities allow us to rewrite the even powers of sine or cosine in terms of the first power of cosine.

For example, using power reducing identities one can easily prove that

$\cos^4 x = \dfrac{1}{8}\cos 4x + \dfrac{1}{2}\cos 2x + \dfrac{3}{8}$ and $\sin^4 x = \dfrac{1}{8}\cos 4x - \dfrac{1}{2}\cos 2x + \dfrac{3}{8}$ (Try it!).

(iii) Power reducing formulas are important in higher level mathematics.

Triple-Angle Identities

Using double angle identities, we can derive triple angle identities.

Identity 3.12: $\sin 3A = 3\sin A - 4\sin^3 A$

Proof.

$$\begin{aligned}\text{We have, } \sin 3A &= \sin(2A + A) = \sin 2A\cos A + \cos 2A \sin A\\ &= 2\sin A\cos^2 A + \left(1 - 2\sin^2 A\right)\sin A\\ &= 2\sin A\left(1 - \sin^2 A\right) + \left(1 - 2\sin^2 A\right)\sin A\\ &= 3\sin A - 4\sin^3 A\end{aligned}$$